8.4 Proportionality Theorems

Geometry 8.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- use a segment parallel to one side of a triangle to find a missing length.
- decide whether a segment is parallel to a side of a triangle.
- use parallel lines and angle bisectors to find missing lengths.

Vocabulary · fill in the missing word

Triangle Proportionality Theorem — a line _____ to one side of a triangle that crosses the other two sides divides them proportionally

Converse — if a line divides two sides of a triangle _____, it is parallel to the third side

Transversal — a line that _____ two or more lines

Angle bisector — a ray that divides an angle into two _____ angles

Part 1 · Discovery: lined paper

Transversals. Draw two slanted lines across at least four lines of your notebook paper. Measure the pieces between the same three ruled lines (in mm).

	Piece 1	Piece 2	Piece 1 ÷ Piece 2
Transversal 1			
Transversal 2			

Triangle. Draw a large triangle ABC with $\overline{BC}$ on a ruled line. Where a ruled line in between crosses the sides, label D (on $\overline{AB}$) and E (on $\overline{AC}$).

AD	DB	AD ÷ DB	AE	EC	AE ÷ EC

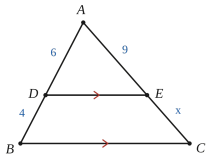
What do you notice? Why do you think it works?

Part 2 · Notes and examples (fill in each note, then work the example)

Triangle Proportionality Theorem: if a line _____ to one side of a triangle intersects the other two sides, then it divides the two sides _____.

Example 1

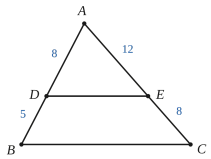
In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. AD = 6, DB = 4, and AE = 9. Find EC.



Converse: if a line divides two sides of a triangle proportionally, then it is _____ to the third side.

Example 2

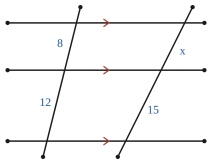
In △ABC, AD = 8, DB = 5, AE = 12, and EC = 8. Is $\overline{DE} \parallel \overline{BC}$?



Three Parallel Lines Theorem: if three parallel lines intersect two _____, then they divide the transversals proportionally.

Example 3

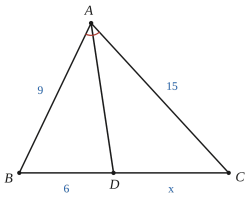
Three parallel lines intersect two transversals. On one transversal the pieces are 12 and 8. On the other, the matching pieces are 15 and x. Find x.



Triangle Angle Bisector Theorem: if a ray bisects an angle of a triangle, then it divides the opposite side into segments proportional to the other two _____.

Example 4

$\overline{AD}$ bisects ∠BAC. AB = 9, AC = 15, and BD = 6. Find DC.



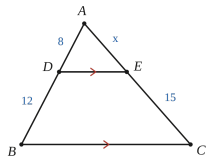
Example 5

Main Street, Oak Street, and Elm Street are parallel. First Avenue crosses them with blocks of 300 feet (Main to Oak) and 240 feet (Oak to Elm). Park Avenue also crosses them; its block from Main to Oak is 400 feet. How long is its block from Oak to Elm?

Part 3 · Partner practice

Partner 1

In △ABC, $\overline{DE} \parallel \overline{BC}$. AD = 8, DB = 12, and EC = 15. Find AE.



Partner 2

$\overline{AD}$ bisects $\angle BAC$. AB = 12, AC = 8, and DC = 6. Find BD.

